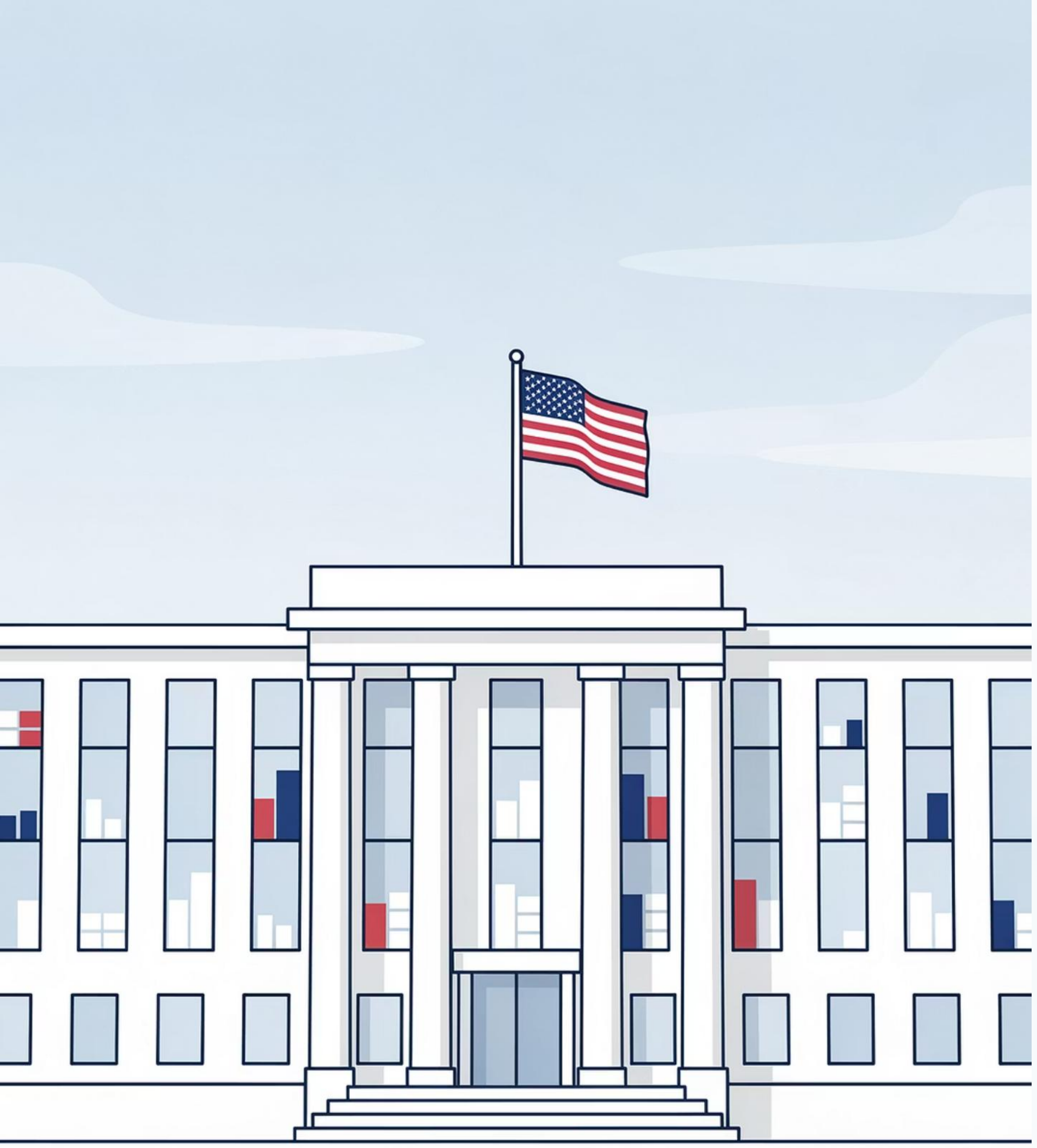
EV Pulse NC: Prioritizing North Carolina's $109M EV Charging Investment

Wolfgang Sanyer │ BIDA 670 — Advanced Analytics
Dr. Majed Al-Ghandour, Advisor
Fayetteville State University · Spring 2026



THE PROBLEM

$109 Million. 100 Counties. No Framework.

North Carolina has received a cumulative **$109M federal NEVI allocation (FY2022–FY2026)**. The decision of where it goes has no validated, county-level scoring methodology — yet.

Federal NEVI Program

FY2022–2026 cumulative formula allocation

100 Counties

All competing for limited infrastructure dollars

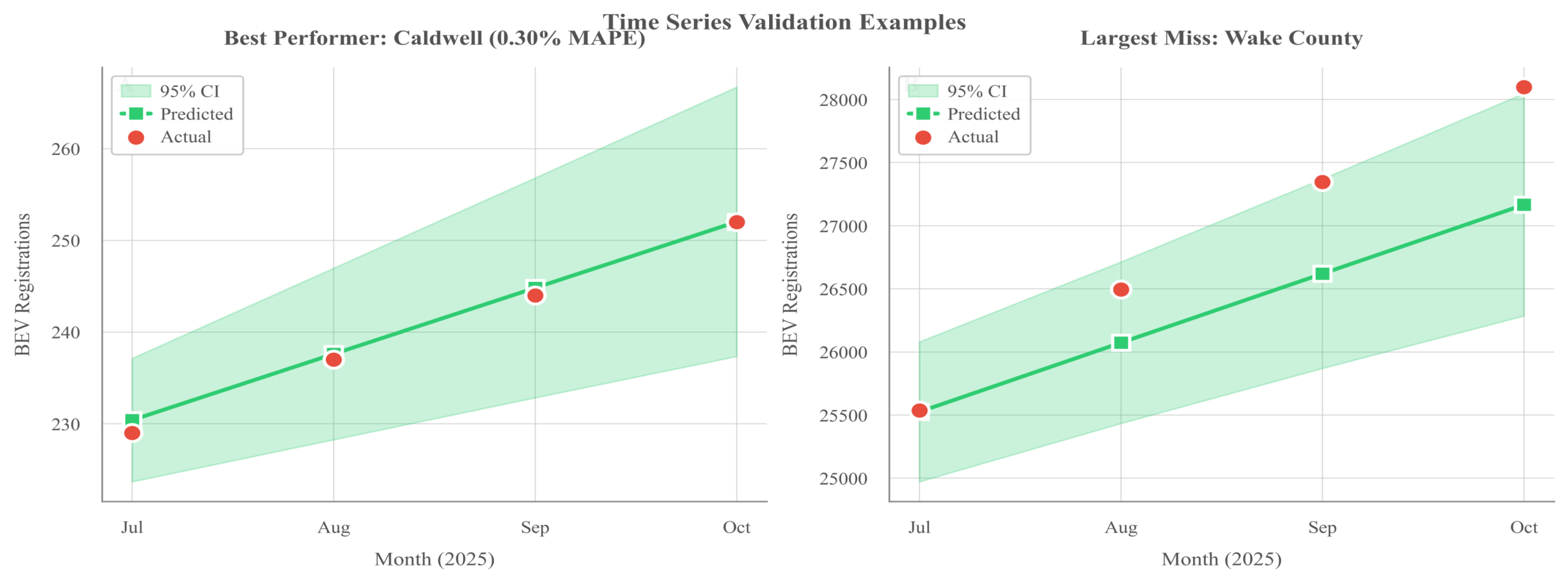
Zero Validated Framework

No existing county-level prioritization model

This is not a hypothetical. This is a live federal allocation decision.

DEMAND SIGNAL

1,727% Fleet Growth in Seven Years



53.8%

CAGR

Sep 2018 – Jun 2025

82

Months

County-level panel data

4.34%

MAPE

400-observation holdout validation

The demand side is clear. The question is whether the supply side is keeping up.

INFRASTRUCTURE GAP

58% of ZIP Codes Have Zero Charging Stations

Statewide Supply

1,985 stations · 6,145 connectors (Feb 2026)

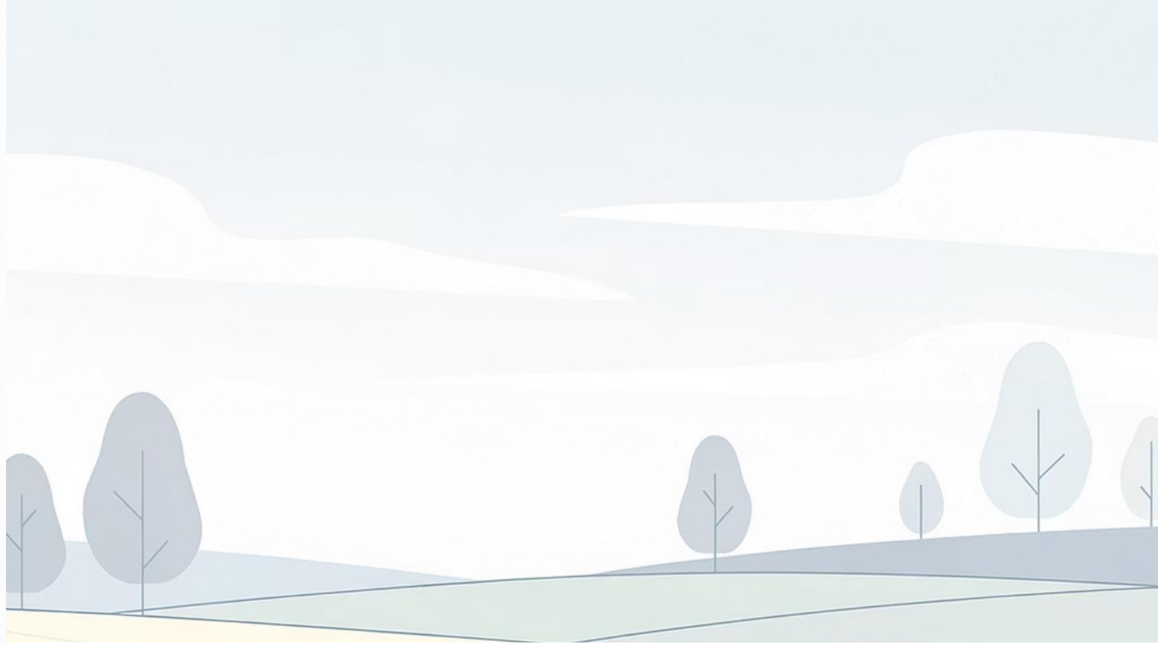
Charging Deserts

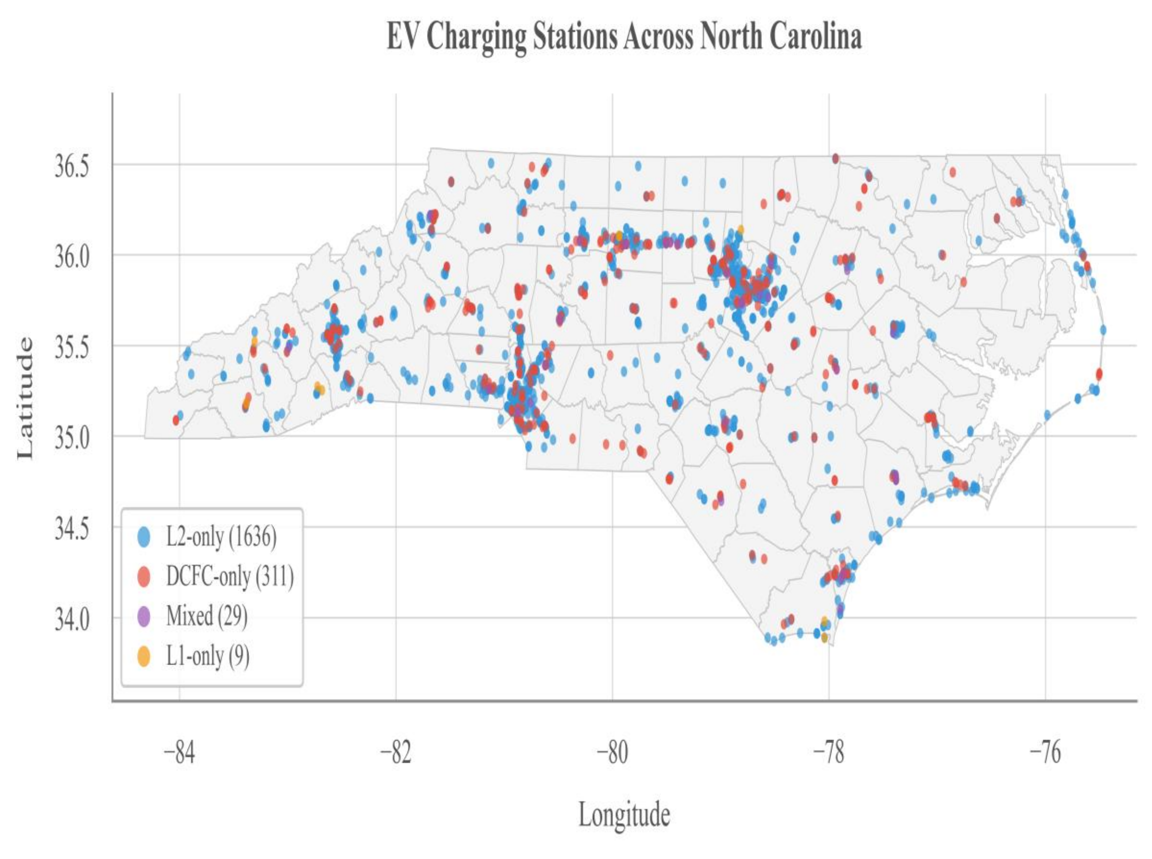
499 of 853 ZCTAs have zero stations — 2.2M residents underserved

Concentrated Growth

47.6% of all infrastructure added in 2024–2025 alone — same locations

⚠ Almost half the infrastructure is brand new — and it's concentrated in the same places that already had it.





EQUITY FINDING

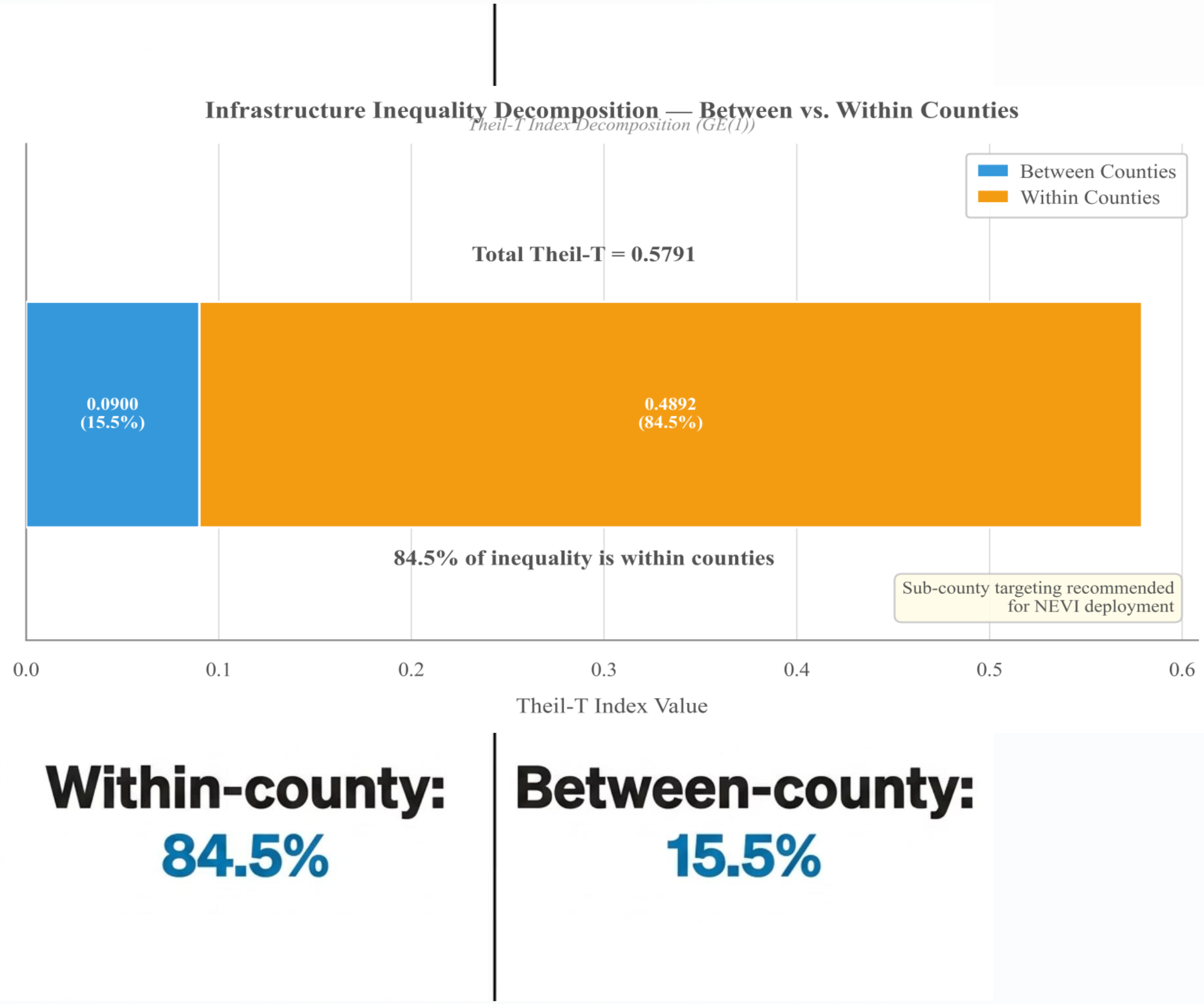
The Inequality Is Inside Counties, Not Between Them

Why This Reframes Everything

A Gini coefficient of **0.805** places NC's EV charging inequality in the extreme range — comparable to severe income inequality.

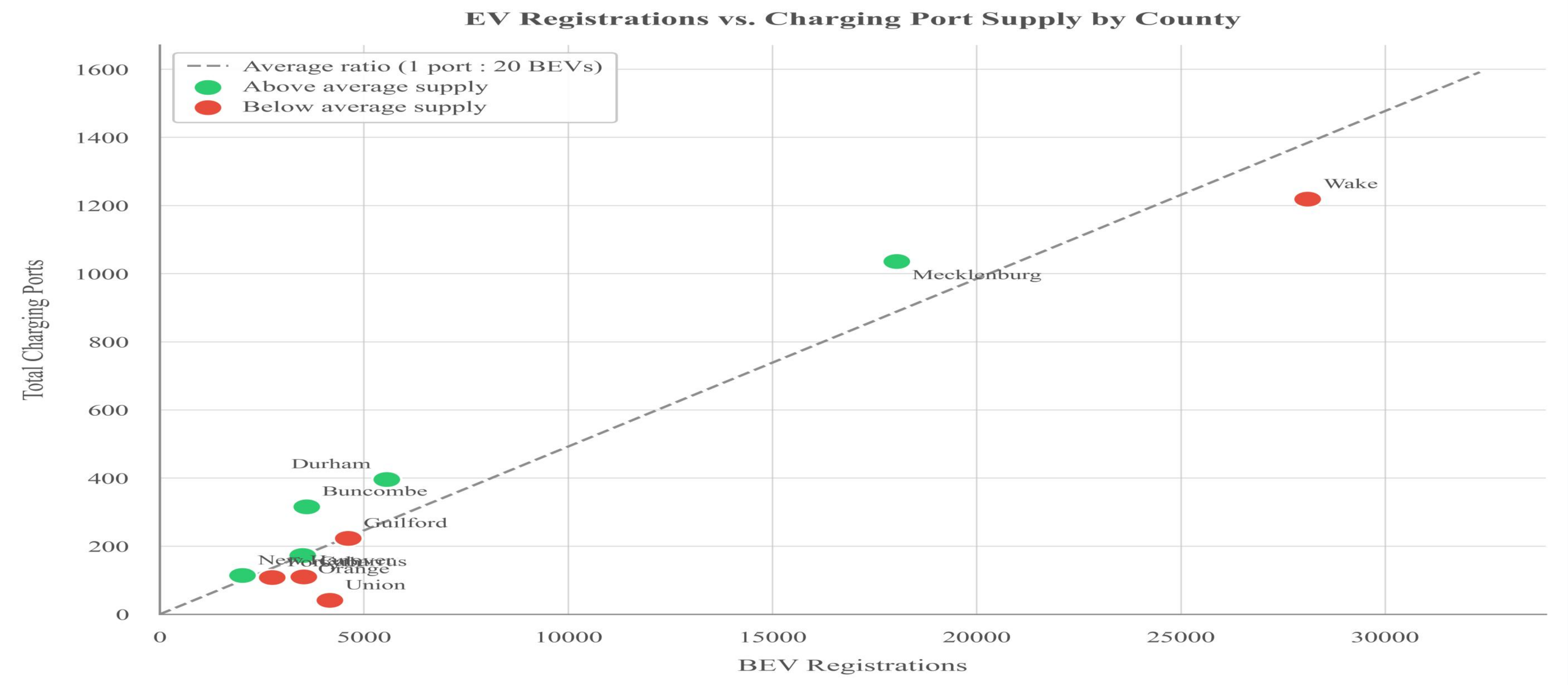
Theil-T decomposition reveals the real fault line: **84.5% of inequality is within counties**. The EV equity problem is not rural vs. urban. It's underserved neighborhoods inside urban counties.

> Policy must target sub-county geographies, not just county boundaries.



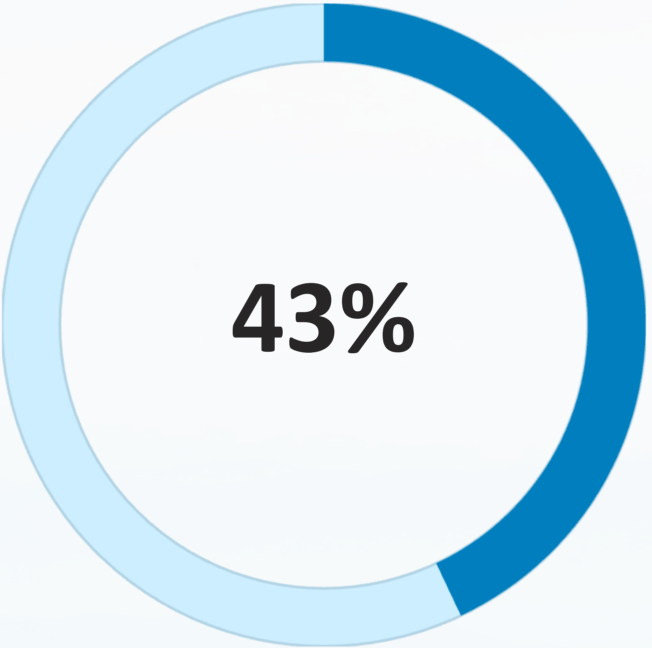
UTILIZATION PRESSURE

More EVs Than Ports — by a Wide Margin



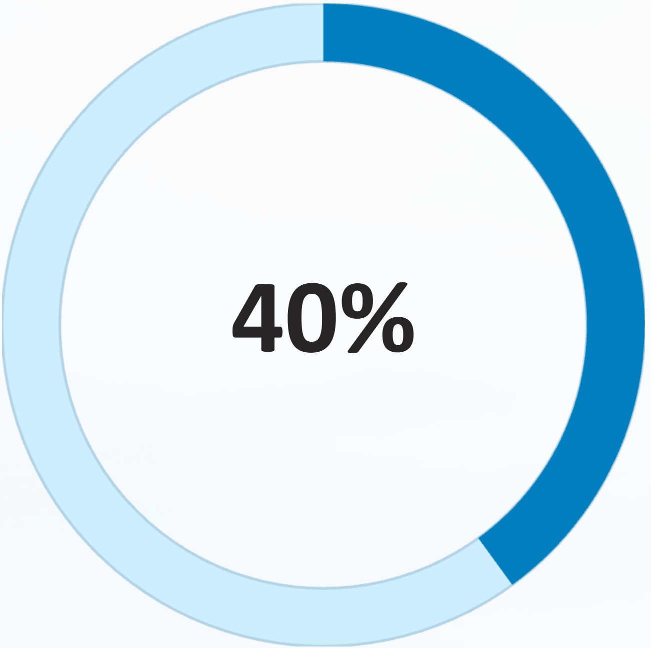
JUSTICE40 MANDATE

Federal Law Requires 40%. Reality Delivers 25%.



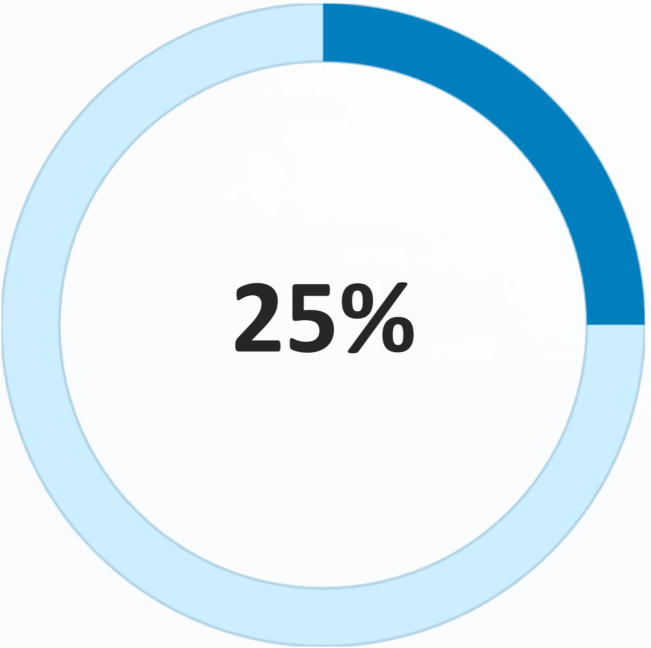
Disadvantaged Tracts

934 of 2,170 evaluable census tracts (CEJST v2.0)



Justice40 Mandate

Federal requirement: 40% of NEVI benefits to disadvantaged communities



Current Reality

Only 24.9% of stations in the study area serve disadvantaged tracts

The federal mandate says 40% of benefits should reach these communities. Currently, less than 25% of stations are there.

THE FRAMEWORK

Three Pillars. Three Federal Mandates. One Score.

Equity
40% weight — Justice40 mandate

NEVI Priority Score

Cost-Effectiveness
25% weight — efficiency

Utilization
35% weight — demand pressure

Statistical validation: **VIF max 1.41** — pillars are independent. Every variable traces to a federal dataset.

Weight Rationale

40% — Equity

Largest weight; directly aligns with Justice40 federal mandate

35% — Utilization

BEV-to-port ratio; captures real-world demand pressure

25% — Cost-Effectiveness

Infrastructure density relative to investment opportunity

THE RANKINGS

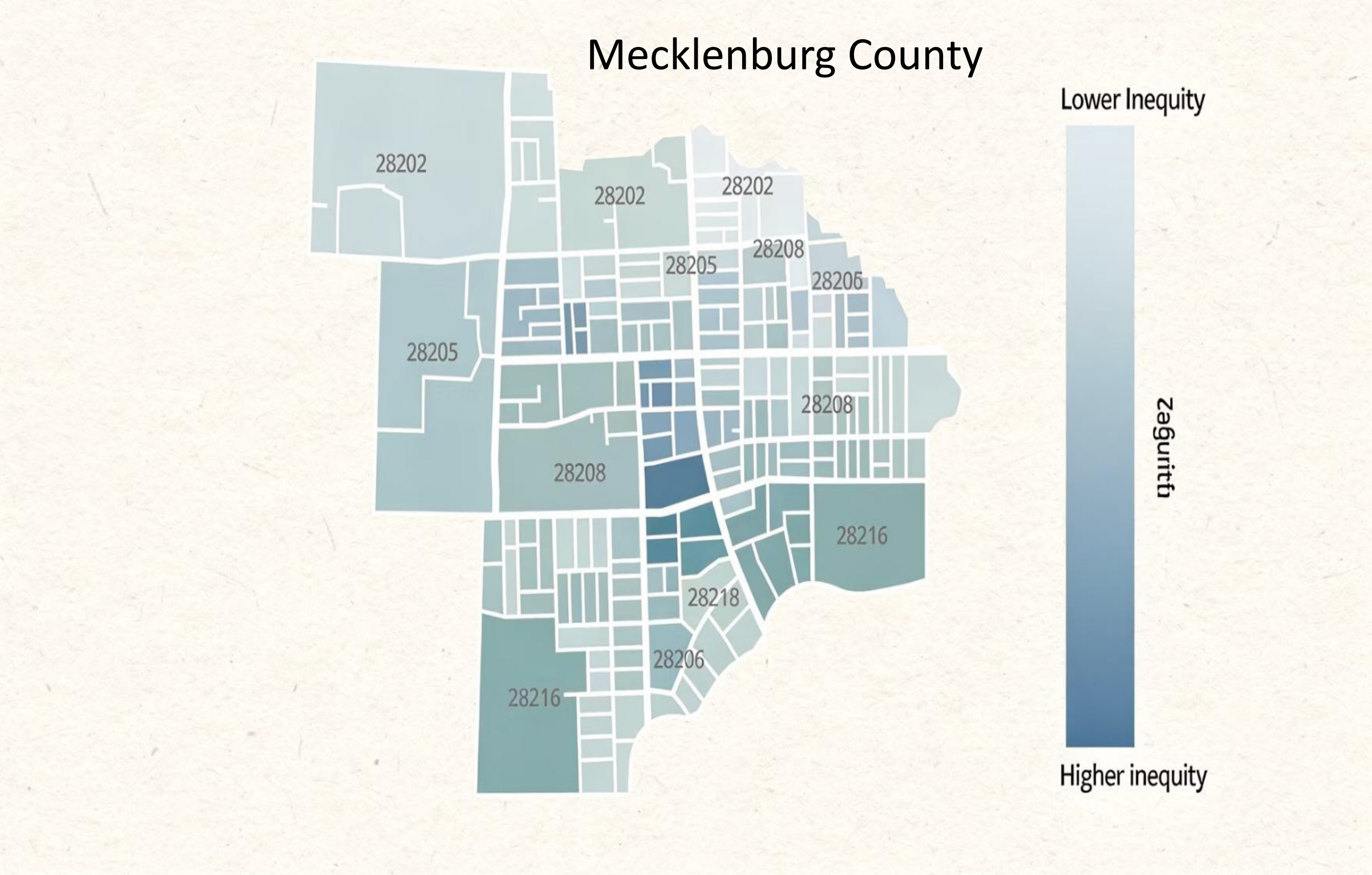
Top 3 Counties — Stable Across Every Sensitivity Test

Rank	County	NEVI Score	Investment Archetype
1	**Union**	**0.561**	Utilization-driven — needs more stations urgently
2	**Mecklenburg**	**0.548**	Equity-driven — needs better-targeted station placement
3	**Guilford**	**0.465**	Balanced across all three pillars
4	New Hanover	0.341	Balanced with equity signal
5	Wake	0.316	High cost-effectiveness; already relatively well-served
6	Durham	0.313	Balanced; moderate equity concerns
7	Forsyth	0.285	Cost-effectiveness driven
8	Cabarrus	0.199	Lower across all pillars
9	Buncombe	0.197	Rural; lower utilization pressure
10	Orange	0.077	Lowest priority — already relatively well-served

These three counties have the most defensible claims for NEVI investment — and each needs a different kind of investment.

SUB-COUNTY TARGETING

County Rankings Tell You Where. ZIP-Level Tells You For Whom.



Two-Tier Architecture

Tier 1: County NEVI Priority Score — allocates federal dollars to the right counties.

Tier 2: ZIP-level equity targeting — directs station placement to underserved neighborhoods within those counties.

> Ranking counties is only half the answer. The real question is which neighborhoods are being left behind.

Top 20 Underserved ZIPs

Identified statewide using composite underservice index

250x Variation

Port density range within Mecklenburg County alone

VALIDATION & RIGOR

Every Number Validated, Traceable, Reproducible

4.34% MAPE

Forecast accuracy on 400-observation true holdout

23 of 23

Federal-standard spatial validation checks passed

VIF Max 1.41

Pillars are statistically independent — no multicollinearity

Top 3 Stable

Rankings unchanged across all sensitivity tests

Zero Open Issues

60-item data quality audit — fully cleared

A policy recommendation is only as strong as the data behind it.

IMPACT & IMPLICATIONS

One Framework. Five Stakeholder Groups. Ready to Deploy.

NEVI Administrators

NCDOT / FHWA — Audit-ready methodology with full data lineage for federal program administrators

County & Regional Planners

Sub-county ZIP-level targeting for local site selection and grant applications

Private Charging Networks

Market intelligence and white-space identification for capital deployment

Academic & Research Community

Reproducible methodology — first mentioned Theil-T decomposition of EV charging inequality

Public & Justice40 Communities

Procedural and distributional fairness — 40% benefits floor for disadvantaged communities

One framework. Five stakeholder groups. One reproducible pipeline. Ready to deploy.

LIMITATIONS & FUTURE WORK

Scope & Clear Path Forward.

Current Scope

- **10-county validation** covers 73% of NC's BEV fleet
- **4-month holdout** window; 12-month extension improves robustness
- **No dollar allocation** — per-station cost modeling required for Phase 6

Roadmap

1

Phase 6

Per-station cost modeling for direct dollar allocation

2

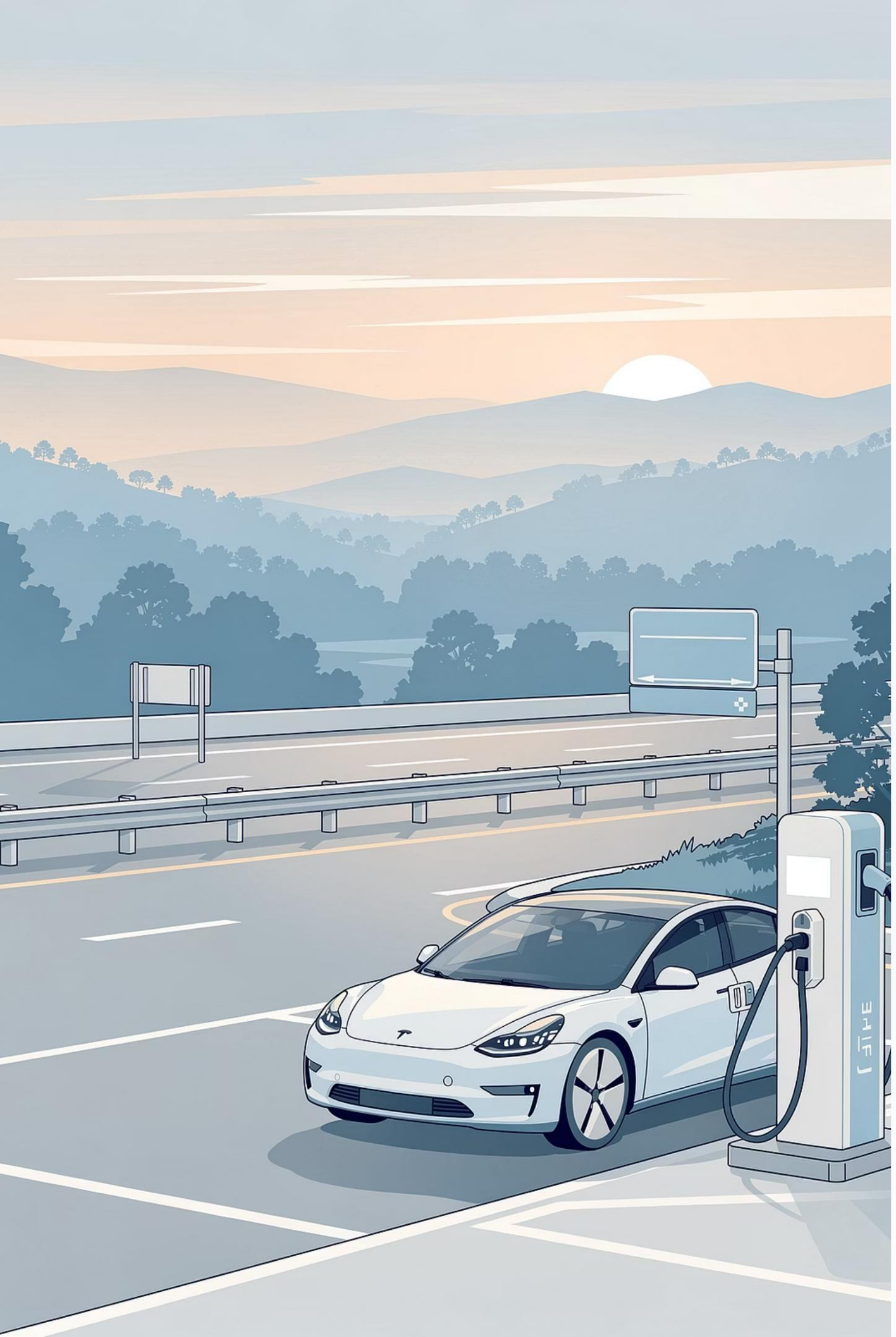
Phase 7

Validate against NCDOT's actual planned NEVI deployments

3

Statewide

Extend framework to all 100 NC counties



Thank You

Wolfgang Sanyer · BIDA 670 — Advanced Analytics · Spring 2026

Email

wsanyer@broncos.uncfsu.edu

GitHub

github.com/wolfieman/ev-pulse-nc

Advisor

Dr. Majed Al-Ghandour

Institution

Fayetteville State University · Broadwell College of Business & Economics